



*"The history of astronomy is a history
of receding horizons."*

Edwin Hubble

OUTSIDE THE BOUNDS of the Milky Way stretch vast gulfs of space, the realm of the galaxies. The closest are on our own galactic doorstep—there is even a small galaxy currently in collision with the Milky Way. The farthest lie billions of light-years away, at the edge of the visible universe—their light has been traveling toward Earth for most of time. Galaxies range from great wheeling disks of matter to giant, diffuse globes of billions of stars and from starless clouds of gas to brilliant furnaces lit up by star formation. They are also violent—despite their stately motion over millions and billions of years, collisions are frequent and spectacular. Collisions disrupt galaxies, sending material spiraling into the supermassive black holes at their centers, fueling activity that may outshine ordinary galaxies many times over. Galaxies influence their surroundings and form constantly evolving clusters and superclusters. At the largest scale, it is these galaxy superclusters that define the structure of the universe itself.

CLOSE ENCOUNTER

Seventy million light-years from Earth, the spiral galaxy NGC 1531 is disrupted by a close encounter with the smaller NGC 1532. The smaller galaxy's gravity disrupts the larger one's spiral arms and distorts NGC 1531's shape, but it also triggers a huge wave of star birth seen in bright star purple-hued star clusters.

BEYOND THE MILKY WAY